FYUGP B.Sc. PHYSICS HONOURS

PROPERTIES OF SOLIDS

Semester V · Major Elective (Materials Science)

MODULE 1: CRYSTAL STRUCTURE

Model questions on topics the past papers miss

Section A: 1 mark each, answer all

Primitive translation vectors

1 Define a primitive translation vector. **(1 mark)**

Point groups and Bravais lattices

2 How many point groups are possible in three-dimensional space? **(1 mark)**

3 How many distinct Bravais lattices are possible in three dimensions? **(1 mark)**

Coordination and the simple cubic structure

4 What is the coordination number of a simple cubic lattice? **(1 mark)**

Lattice directions

5 How many members does the family of face diagonals <110> of a cube have? **(1 mark)**

6 How many members does the family of body diagonals <111> of a cube have? **(1 mark)**

Section B: 2 marks each, answer all

Symmetry operations

7 State the permitted values of n for an n-fold rotation axis of a lattice. **(2 marks)**

8 Name the two-dimensional point groups of the rectangular crystal system. **(2 marks)**

Unit cell and basis

9 Find the effective number of lattice points in the conventional cell of a body-centred cubic lattice. **(2 marks)**

10 State the basis of the sodium chloride structure with respect to a face-centred cubic lattice. **(2 marks)**

Section C: 4 marks each, answer all

Angles between planes and reciprocal lattice

11 Find the angle between the normals to the (100) and (110) planes of a cubic crystal. **(4 marks)**

Reciprocal lattice

12 Obtain the reciprocal lattice vectors of a simple cubic lattice of lattice constant a. **(4 marks)**

Packing fraction

13 Find the packing fraction of a simple cubic structure. **(4 marks)**

Zinc blende structure

14 Describe the zinc blende structure. State its lattice and basis, its coordination number, and the nearest-neighbour distance in terms of the lattice constant a. **(4 marks)**

Brillouin zones

15 Show that the first Brillouin zone of a two-dimensional square lattice of side a is a square of side $2\pi/a$. **(4 marks)**

Simple cubic structure

16 A simple cubic lattice has lattice constant a. Find the number of nearest neighbours of an atom and the nearest-neighbour distance. **(4 marks)**

Section D: 10 marks each, answer any one

Sodium chloride structure

17 Describe the sodium chloride structure as a face-centred cubic lattice with a two-ion basis. Give the positions of the Na^{+} and Cl^{-} ions in the unit cell, state the number of molecules per unit cell, and show that the lattice constant of touching ions of radii r_{+} and r_{-} is $a = 2(r_{+} + r_{-})$. **(10 marks)**

Symmetry operations

18 Show that a lattice cannot have a five-fold rotation axis. List the permitted rotation axes, and list the five distinct two-dimensional Bravais lattices. **(10 marks)**

MODULE 1: MODEL ANSWERS

For checking your work. Each answer shows the key step and the result.

Section A. 1. A translation vector is $\mathbf{T} = n_1\mathbf{a} + n_2\mathbf{b} + n_3\mathbf{c}$. It is primitive when the coefficients n_1, n_2, n_3 are integers (book §1.2). **2.** 32 point groups (book §1.6). **3.** 14 Bravais lattices (book §1.7). **4.** 6 (book §1.10.2: each atom touches six neighbours along the cube edges). **5.** 12 members (book §1.8). **6.** 8 members (book §1.8).

Section B. 7. n = 1, 2, 3, 4 or 6 (book §1.5, Eqs. 1.3 and 1.4). **8.** 1m and 2mm (book Table 1.1). **9.** $N = N_i + N_f/2 + N_c/8 = 1 + 0 + 8/8 = 2$ (book §1.7, Eq. 1.5: the body-centre point is inside the cell and counts 1; each of the eight corners counts 1/8). **10.** Na^+ at (0, 0, 0) and Cl^- at (a/2, 0, 0) of an fcc lattice (book §1.13).

Section C. 11. Normal to (100) is [100]; normal to (110) is [110]. $\cos\theta = 1/\sqrt{2}$, so $\theta = 45°$ (book §1.8, Eq. 1.6). **12.** Reciprocal lattice of a simple cubic lattice is simple cubic: $\mathbf{a}^* = (2\pi/a)\mathbf{i}$, $\mathbf{b}^* = (2\pi/a)\mathbf{j}$, $\mathbf{c}^* = (2\pi/a)\mathbf{k}$ (book §2.4.2, Eq. 2.21). **13.** Simple cubic: $a = 2r$, one atom per cell, so $f = (4/3)\pi r^3/a^3 = \pi/6 \approx 0.52$ (book §1.10.2). **14.** An fcc lattice with a two-atom basis: one atom at (0, 0, 0) and one at (a/4)(1, 1, 1). Coordination number 4; nearest-neighbour distance $\sqrt{3}a/4$ (book §1.11 and §1.12). **15.** The condition from the Ewald-type relation (Eq. 2.31) for $h = \pm1$, $k = 0$ gives $k_x = \pm\pi/a$, and for $h = 0$, $k = \pm1$ gives $k_y = \pm\pi/a$. These four lines bound a square of side $2\pi/a$ (book §2.7, Fig. 2.14). **16.** 6 nearest neighbours at distance a (book Problem 5 of chapter 1, printed answer 6, a).

Section D. 17. Na^+ at (0,0,0), (½,½,0), (½,0,½), (0,½,½); Cl^- at (½,½,½), (0,0,½), (0,½,0), (½,0,0). Four molecules per cell. Along a cube edge the ions touch: $a = 2(r_+ + r_-)$ (book §1.13 and Example 1.5). **18.** Lattice translation requires $\cos\theta = (m - 1)/2$ with $|\cos\theta| \le 1$, so m = 3, 2, 1, 0, −1. This gives θ = 0°, 60°, 90°, 120°, 180°, and n = 1, 6, 4, 3, 2 (book §1.5, Eq. 1.4). Five is not allowed. The five two-dimensional Bravais lattices are oblique, rectangular primitive, rectangular centred, square and hexagonal (book §1.7 and Table 1.1).

Mapping to the syllabus and textbook

Syllabus item	Topic	Puri and Babbar section	Model questions
1	Crystal lattice, translation vectors, unit cell and basis	§1.2, §1.3, §1.4	1
2	Symmetry operations, point groups and space groups	§1.5, §1.6	2, 7, 8, 18
3	Bravais lattices, lattice directions and planes, interplanar spacing	§1.7, §1.8, §1.9	3, 5, 6, 11, 18
4	Simple crystal structures with examples	§1.10 to §1.13	4, 9, 10, 13, 14, 16, 17
5	X-ray diffraction, reciprocal lattice, Brillouin zones	§2.4, §2.7	12, 15

Questions are drafted for the gaps in the past papers (not from any past paper). Section 2.3 (X-ray diffraction methods) is outside the Module I scope, so no question uses it.